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WHITE PAPER · 2026

Practice Before Patients

What the research on Virteasy haptic simulation means for learning pathways and operational productivity in dental schools

Edition for the Deans attending the ADEA Deans' Meeting

Saratoga · Exclusive distributor of Virteasy Dental for Italy, Spain, North America and the Middle East

EXECUTIVE SUMMARY

Simulation as an added practice layer, not a replacement

Virtual haptic simulation is not a replacement for the phantom lab or the clinic. Across 18 studies and two reviews on Virteasy [1–21] and in five other papers [26–30], it works best as a **additional pre-clinical practice layer**: students repeat procedures at any hour, with objective data on every attempt, before they touch a phantom head or a patient.

Learning pathway

Randomized and comparative studies report faster or better early skill, and lower stress and anxiety, when virtual simulator practice comes first [1–6]. On real patients: higher clinical scores ($p < 0.0001$; 30 students) [26]

Productivity

Objective scoring is documented. Gains in instructor time, consumables and lab scheduling are plausible but **not yet measured** in controlled studies. A pilot can measure them. [2]

249 simulators in 70 institutions

28 countries with installations

22 studies reviewed here

3 systematic and scoping reviews

What deans and program directors can take away

1. Treat the virtual simulator as an added practice step with entry and exit criteria for each skill, not as a substitute for phantom hours.
2. Prefer the headset-free set-up for complex tasks such as access cavity preparation.
3. Budget for faculty calibration and for a transition period back to conventional simulators.
4. Measure before scaling: use a 90-day pilot with KPIs for instructor hours, lab occupancy, consumables and time to competence.

Sources: numbered references in square brackets are listed on page 14. Installation figures: list supplied for this paper (page 4).

THE LEARNING PATHWAY

Where simulation fits between theory and the clinic

A dental curriculum moves students from knowledge to supervised patient care. The research positions the virtual simulator in the middle of that path, between theory and phantom work.

1	**Orientation** Anatomy, instruments and ergonomics on screen.	Students and staff rate it useful (127 and 865 respondents) [16, 17]
2	**Fundamentals** Repeated drills in implant, cavity and access preparation.	Better or faster early results (88 and 40 students) [2, 4]
3	**Feedback** Objective scores and expert review of each attempt.	Agrees with teachers 71 to 78% of the time [12]
4	**Transfer** Phantom-head work, with a planned transition period.	Haptic scores did not predict phantom results [9]
5	**Clinic** Supervised patient care once the skill is shown.	Higher clinical performance in one 30-student study [26]; case reports only for patients [15]

Design principle. Move students on by demonstrated competence, not hours logged. A 2025 Sheffield thesis [22] adds that breaking a task into parts can help, and that motor-skills research does not support replacing the phantom head with VR. A 2017 thesis from Nancy [23] adds that performance stalled or fell once the angulation aid was removed after the fourth attempt, so aids should be withdrawn gradually, with briefing and debriefing.

Sources: references 2, 4, 9, 12, 15, 16, 17, 22, 23 and 26 (page 14).

ADOPTION

A platform already used in dental schools worldwide

The Virteasy installation list includes 249 simulators in 70 institutions and 28 countries between 2015 and 2026.
In Italy, the Dental School of Turin has installed 4 Virteasy Pro units, which have enabled 4 scientific publications.



Universities and schools with Virteasy installations

Source: Virteasy installation list supplied for this paper, 2015 to 2025; see Methods on page 14. Institutions shown appear in that list.

THE EVIDENCE · PART 1

Controlled studies: promising early skill, to be explored further

Nine controlled or comparative studies. They differ in design and topic, so read them as a map of signals, not as pooled proof.

STUDY	DESIGN	SAMPLE	SIGNAL	TAKEAWAY
Joseph, 2014 [1]	Randomized; implant drilling, simulator vs lecture only vs expert dentists	60 (20/20/20)	Positive	Simulator-trained students improved over 8 trials and moved toward expert level. Resin-model perforations 10% vs 20%, not formally tested.
Vincent, 2020 [2]	Randomized; cavity preparation on Virteasy vs plastic teeth, final test on plastic	88 first-year	Positive	Both groups improved and final results were similar. The authors cite savings in supervision time and materials.
Vincent, 2022 [3]	Randomized; implant site with vs without virtual aids	88 first-year	Positive	Students with aids got better results faster; repeated practice built skill without aids.
Usta, 2025 (Int Endod J) [4]	Randomized; VR first vs artificial teeth first	40	Positive	VR first gave higher dexterity and self-rated proficiency and lower anxiety. Stress fell in both groups.
Usta, 2025 (J Endod) [5]	RCT, 3 arms: simulator, mobile app, control	30	Positive	Reported better clinical performance, lower stress and higher confidence at first clinical procedures.
Al Ali, 2024 [6]	Crossover; 3D vs 2D vision on the simulator	24 (21 finished)	Positive	3D vision improved accuracy ($p = 0.035$) and reduced over-cutting. Time did not differ.
Bandiaky, 2026 [8]	Crossover; access cavity, with vs without VR headset	90	Caution	Better haptic parameters without the headset. Most students preferred the non-immersive set-up.
Yildirim-Manav, 2026 [10]	Randomized, 3 groups: VR haptic, CaviPrep plate, control	75	Caution	The plate group did better on outline, retention form and width.

Signal: Positive = favors simulator practice; Caution = result limits use in the applications considered.

Sources: references 1 to 6, 8, 10 and 11 (page 14). Numbers in brackets identify each study.

THE EVIDENCE · PART 2

Feedback, feasibility and what students say

Nine studies on assessment, feasibility and student perception. They show acceptance and workflow potential, and they also show where the experience falls short.

STUDY	DESIGN	SAMPLE	SIGNAL	TAKEAWAY
Dixon, 2020 [12]	Validity test; simulator feedback vs 12 teachers, 10 printed cavities	12 teachers	Positive	Simulator and teachers agreed 71% (pooled) to 78% (modal) of the time. One exercise only.
Carpegna, 2023 [14]	Feasibility; endodontic microsurgery built from patient scans	Not reported	Feasibility	Virtual patient cases give reviewable feedback and saved progress.
Pasqualini, 2025 [15]	Case report; 10 virtual sessions, then supervised surgery	1 resident	Feasibility	Resident reached target scores, then operated; the lesion healed completely at 1 year. One case only.
Towers, 2021 [13]	Interviews; VR then 3D-printed patient-specific model	Not stated	Perception	Students felt more confident and prepared before treating their patient.
Nassief, 2024 [7]	Qualitative; 2D vs 3D vision	Not stated	Perception	Students valued 3D and saw VR as an early adjunct to the phantom head.
Fernandez-Sagredo, 2016 [16]	Cross-sectional survey, Chile	127	Perception	94% positive for teaching manual skills, 91% for autonomous practice, 72% for assessment.
ADEMA survey, Spain [17]	Exploratory survey and interviews	865	Perception	86.7% wanted this earlier in training. Authors are affiliated with the adopting school.
Faid, 2026 [18]	Cross-sectional survey; pediatric pulpotomy	115	Mixed	63.6% satisfied, but 52.7% found it difficult to use (ergonomics, tactile realism).
Bandiaky, 2025 (J Dent Sci) [9]	Prospective pilot; haptic exercises vs crowns on phantom	Not stated	Mixed	Haptic performance index did not correlate with conventional results; a transition period is needed.

Sources: references 7, 9 and 12 to 18 (page 14). Numbers in brackets identify each study.

THE EVIDENCE · PART 3

New studies: from simulator practice to clinical performance

Five recent publications, added in this edition, follow the learning pathway through to performance on real patients. They do not all use Virteasy: the simulator is stated for each study.

STUDY	DESIGN	SAMPLE	SIGNAL	SUMMARY
Sánchez-Herrera, 2026 [26]	Controlled, parallel groups; access cavities on real patients after simulator practice (Simodont) vs direct procedure	30 (15/15), 5th year	Positive	Overall clinical score 8.76 vs 7.31 out of 10 ($p < 0.0001$), blinded raters. Larger drop in anxiety ($p = 0.052$). 86.7% prefer simulator plus artificial teeth.
Folguera, 2026 [27]	Randomized; haptic simulator (SIMtoCARE Dente) vs 3D-printed teeth; 6 access cavities	22 (12/10)	Positive	Only the simulator group significantly improved self-assessment ($p = 0.009$); the between-group difference was not significant. Agreement with experts still low: feedback and guidance are needed.
Ba-Hattab, 2023 [28]	Questionnaire after access cavities on a phantom and on a simulator (VRDS in Qatar, Simodont in Turkey)	60	Perception	76% find it useful for fine motor skills; 85% support using it alongside phantom practice, with improvements needed.
Reymus, 2020 [29]	Questionnaire; root canal anatomy with VR (non-haptic), CBCT and radiographs on extracted teeth	42, 3rd year	Perception	VR was judged better than CBCT and radiographs for understanding canal anatomy (p from < 0.001 to 0.049) and was well accepted.
Moussa, 2022 [30]	Systematic review (PRISMA) on VR, augmented reality and haptic simulation	73 publications	Review	52 of 73 studies show significant improvement in educational outcomes; positive perception. The authors call for larger samples and long-term clinical studies.

8.76 vs 7.31
clinical score on real patients [26]

p = 0.009
more accurate self-assessment [27]

52 of 73
studies with better outcomes [30]

Four context readings [31–34]

Not efficacy studies, but they converge: coaching supports self-efficacy (Fox); shared standards for training and assessment are needed (Azim; Longridge et al.); entrustable professional activities (EPAs) help decide when a student is ready (Kelly).

Sources: references 26 to 34 (page 14). Numbers in square brackets identify each study.

THE EVIDENCE · PART 4

Three benefits reported by faculty: what the sources say

Faculty who use the haptic simulator report three benefits: better-prepared students, less tutor time, fewer materials. We re-read the available articles and Prof. Pasqualini's slides (BU 2026) to check how well each one is documented.

Student performance

Documented, with limits

With Virteasy, simulator practice before work on artificial teeth gives about 10 more points in the practical test and less stress (VAS 3.41 vs 5.92) [4].

With Virteasy, clinical score 80.5 vs 70.6 for controls ($P < 0.05$) and shorter drilling time [5].

On real patients: 8.76 vs 7.31 out of 10 ($p < 0.0001$), with a non-Virteasy simulator [26].

52 of 73 studies show better outcomes [30].

Limits. Small samples, short-term outcomes, no patient outcomes. The value lies in time gained before the patient, not in a substitute for the instructor.

Tutor time

One measured, dated figure

Measured with DentSim (2004): 16 faculty-student interactions vs 42, of 1.9 vs 4.0 minutes; that is 0.5 vs 2.8 faculty hours per student [37].

With Virteasy, the authors cite less supervision time [2]; the review [21] supports this across seven papers.

Limits. A single quantitative figure, from 2004 and another system; none with Virteasy. Instructor feedback remains the best [21, 30].

Consumables

Stated, not measured

Exercises can be repeated at no cost for materials and without plastic teeth to replace [28]; once installed, it is used without physical materials, unlike 3D-printed models [15].

Extracted teeth pose availability and ethical limits [27]. Prof. Pasqualini compares 3D-printed models, cadaver and simulator qualitatively (BU 2026).

Prof. Pasqualini's comparison (BU 2026)

CRITERION	3D-PRINTED MODELS	CADAVER LAB	HAPTIC SIMULATOR
Patient-specific	Yes	No	**Yes, from CBCT/STL**
Repetitions	Limited	Single use	**Unlimited**
Long-term cost	Costly reprint	Very high	**Lower (one-time)**
Risk	Not applicable	Biohazard	**None**

Qualitative assessment presented by the author, not a measurement.

Sources: references 2, 4, 5, 15, 21, 26, 27, 28, 30, 37 and 38 (page 14) and Prof. Pasqualini's presentation, BU 2026. Studies 26 and 27 do not use Virteasy.

THE EVIDENCE · PART 5

Dose, sequence and where simulation does more

A literature review compiled for ADEE [35] examined 80 sources (2002–2026) on simulation versus traditional lab training. It is a synthesis assisted by automated search tools and, for many studies, relies on abstracts: figures should be checked against primary sources. Its conclusions confirm the line of this paper: simulation sits alongside the lab, it does not replace it.

74.3%
of 35 randomized trials (about 2,100 students) with a significant improvement in at least one outcome [36]

6–10 hours
of exposure: superiority emerges with doses beyond a few hours [35]

80.7 vs 57.8
knowledge at one year in orthognathic surgery ($p < 0.0001$) [35]

WHERE	SIMULATION DOES MORE	THE LAB REMAINS BETTER
Type of task	Spatial orientation and complex anatomy: endodontic access, radiology, orthognathic surgery, implantology.	Fine tactile discrimination: veneer margins, finger rest in scaling, local anesthesia.
Experience	Students with manual basics already acquired.	Absolute beginners in complex preparations: in the crown study, worse with simulation ($p < 0.0001$).
Combination	Simulation plus physical practice beats either alone in periodontology, implantology, prosthodontics, endodontics and restorative dentistry.	—

How much time: dose matters

EXPOSURE	OUTCOME	SOURCE
60 minutes	Equal to the traditional method ($p = 0.315$): endodontic access	Slaczka 2024, in [35]
3 hours (self-directed)	No improvement in crown preparation ($p = 0.24$)	Odisho 2025, in [35]
6–10 hours	Better trajectory ($p = 0.01$); at 8 hours, advantage on 2 of 3 exams	LeBlanc 2004; Urbankova 2010, in [35]

Simulation before or after?

In endodontics, haptic simulation before artificial teeth gives more dexterity ($p < 0.05$) and less stress [4]; in periodontology the physical model first works better, and in other studies the order does not change results. The effect depends on the procedure.

Sources: references 4, 35, 36 and 39 (page 14); individual study data are taken from the review and were not re-read in the primary sources.

THE EVIDENCE · PART 6

How to integrate simulation: guidance and resources

The same review and the ADEE infographic "Bridging the Gap" translate the evidence into guidance for schools. We report them as a summary reading, with the figures available.

1 · Simulation first

Introducing it before physical practice speeds learning and reduces anxiety, especially in complex procedures [4, 26, 35].

2 · A dose of 4–10 hours

A few hours tend not to beat the traditional method: plan a volume of practice per skill [35].

3 · Automated assessment

Objective, repeatable scores reduce subjectivity: 93.3% of students in the digital group judge it objective vs 73.3% [35].

Simulation: best for

High spatial complexity and anatomical visualization: endodontics and radiology.

Lab: best for

Tactile sensitivity and fine proprioception: veneers and scaling.

Hybrid: for first skills

Simulation builds the mental model; the lab builds motor templates.

The resources to put on the table

ITEM	REPORTED FIGURE	HOW TO READ IT
Faculty time	0.5 vs 2.8 hours per student; 16 vs 42 interactions [37]	Measured in 2004 with DentSim: shows the direction, not an estimate for Virteasy.
Faculty training	About $5,000 per instructor; 40–60 hours to master the interfaces [38]	Often underestimated; affects adoption.
Assessment	Simulator scores predict faculty grades with 82.7% accuracy [35]	Single-group study; to be validated.

What follows

A block-based pathway: virtual simulation, then lab, then patient, with indicators measured by the institution (page 12).

Sources: references 35, 37 and 38 (page 14); ADEE infographic "Integrating Simulation in Dental Education". Cost figures are generic and not specific to Virteasy.

WHAT USERS REPORT

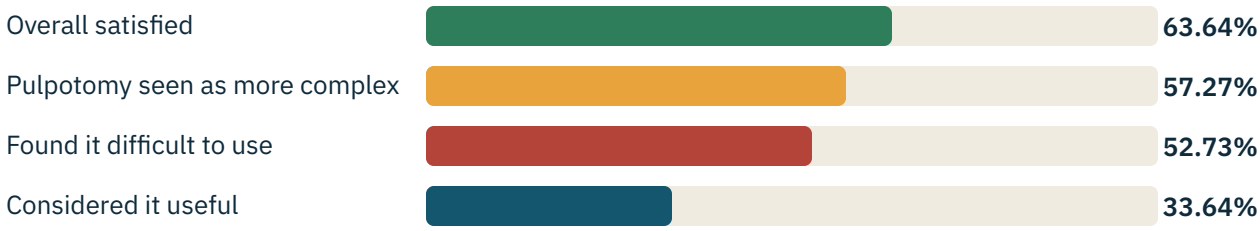
Strong on teaching and practice, weaker on realism

Two surveys give the clearest view of how users judge the simulator. Teaching and self-practice score highest; realism and ease of use score lowest.

Chile survey, 127 students, dentists and faculty: share positive [16]



Morocco survey, 115 students, pediatric dentistry: share of students [18]



+3D

Stereoscopic vision improved accuracy (p = 0.035) with no difference in task time. [6]

-Headset

Performance was better without the VR headset in 90 third-year students. [8]

1 to 4

Studies agree that practice should be added to, not replace, phantom work. [2, 7, 9, 22]

Percentages as reported in each source.

Sources: references 2, 6, 7, 8, 9, 16, 18 and 22 (page 14).

OPERATIONS LENS

Where productivity could come from

For those who run teaching and course management, seven operational levers are worth testing. Only objective assessment data is directly documented.

Assessment data	Objective metrics and saved progress for every attempt	Documented
Time to competence	Vincent 2020 [2]: equal final results; a review [19–21] cites 22% shorter procedure time (543 to 424 s). 3D vision showed no time effect [6].	Early signal
Instructor time	Measured with another system (DentSim, 2004): 0.5 vs 2.8 faculty hours per student (ref. 37). Vincent 2020 authors cite saved supervision time; it was not measured as an outcome [2]	Early signal
Consumables	A 2021 review [20] argues phantom materials and waste are recurring costs; a 2017 thesis notes exercises can be repeated without extra cost (author statement, not measured) [23]	Opportunity
Lab scheduling	Practice available outside supervised lab slots	Opportunity
Case build time	A new patient-specific scenario took weeks to months to create (new avatar 2 to 4 months; a further case about 2 weeks) [13–15]	Documented

Pilot KPIs

- Instructor hours per student per skill
- Lab occupancy and idle hours
- Consumables per student
- Time to competence
- Students needing remediation

Sources: references 2, 6, 13 to 15, 19 to 21, 23 and 37 (page 14). Pilot KPIs are a Saratoga proposal, not a finding.

NEXT STEP

A 90-day pilot to measure what matters

A short, structured pilot turns assumptions into local numbers.

Days 1 to 15

Baseline

Record current hours, lab use, consumables and skill scores.

Days 16 to 30

Set-up

Install simulators, calibrate faculty, define entry and exit criteria.

Days 31 to 75

Run

Run one cohort with simulator practice and one with the usual curriculum.

Days 76 to 90

Review

Compare KPIs, collect student feedback, decide on scale-up.

ABOUT SARATOGA

Saratoga is the exclusive distributor of Virteasy Dental in Italy, Spain, North America and the Middle East. Based in Pordenone, Italy, Saratoga also manufactures dental simulators and develops EasyTeachCloud, an e-learning platform that supports students' learning pathway. Its latest research project, ASISME (Advanced Simulation Student Monitoring Ecosystem), combines artificial intelligence, computer vision, blockchain and a digital twin to record how students interact with the simulators, assess their performance objectively and evaluate their working posture.

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Sources: the 90-day plan is a Saratoga proposal, not drawn from the cited studies. Company information supplied by Saratoga.

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Sources and how this paper was built

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Methods and notes

Sources come from the research list supplied for this paper and were consulted as follows: full text for references 4, 5, 8, 14, 15, 21 and 26 to 34, plus Prof. Pasqualini's slides (BU 2026); abstracts or secondary sources for the others. Summaries are paraphrased. The 2019 scoping review declares a research collaboration with the Virteasy developer. The haptic algorithm behind Virteasy was co-invented by the author of ref 25, and the developer is Didhaptic (refs 24, 25). Productivity levers are hypotheses to be tested in a pilot, not findings. References 31–34 are perspectives and narrative reviews. References 35–39 come from the ADEE review (search assisted by automated tools): data from studies 36–39 are cited second-hand and should be verified in the primary sources.

Numbers in square brackets throughout this paper refer to this list.